

Pre-Snap Motion as a Predictive Indicator for Play-Calling

BRETT BARNES

Key Definitions:

Motion: movement of an offensive player initiated after being set ('lined up') but before the snap, where the player is *still moving at the time of the snap*

Personnel: the amount and types of players that are on the field for a given play, denoted by a two digit number representing the number of running backs and number of tight ends on the field

For example: a play with two running backs and one tight end would have a 21 personnel group

Formation: the pattern the players are physically arranged in *after* offensive shifts yet *before* motion

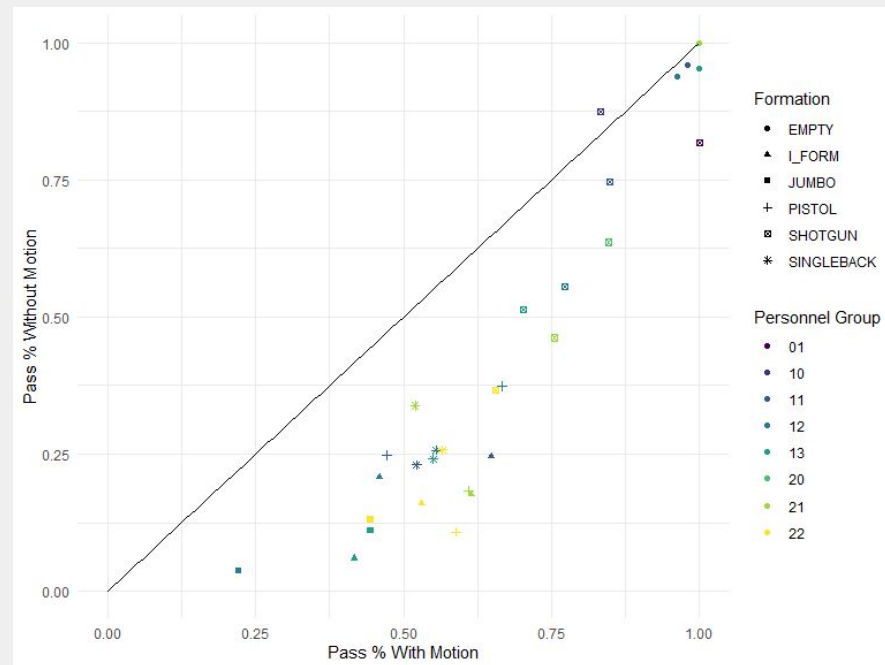
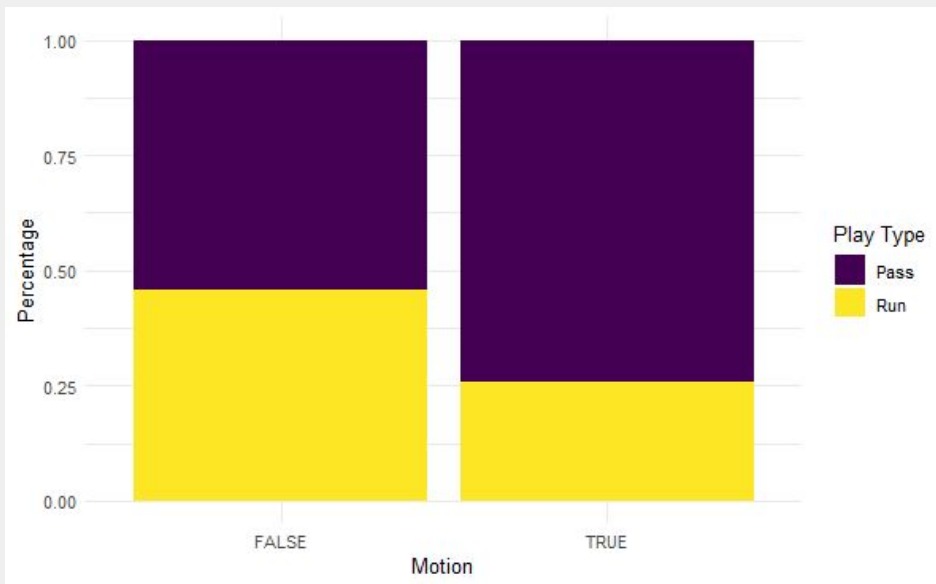
The Dataset:

NFL Big Data Bowl 2025: 'Help use pre-snap behavior to predict and better understand NFL team and player tendencies'

Composed of ~6.5 million rows and >100 columns across 13 tables, the most critical tables I used was a table of each play from each player's perspective, a table of each player's statistics, and a table of each play that occurred and its results (e.g. play type, yards gained, time to throw, defensive coverage, EPA, etc.)

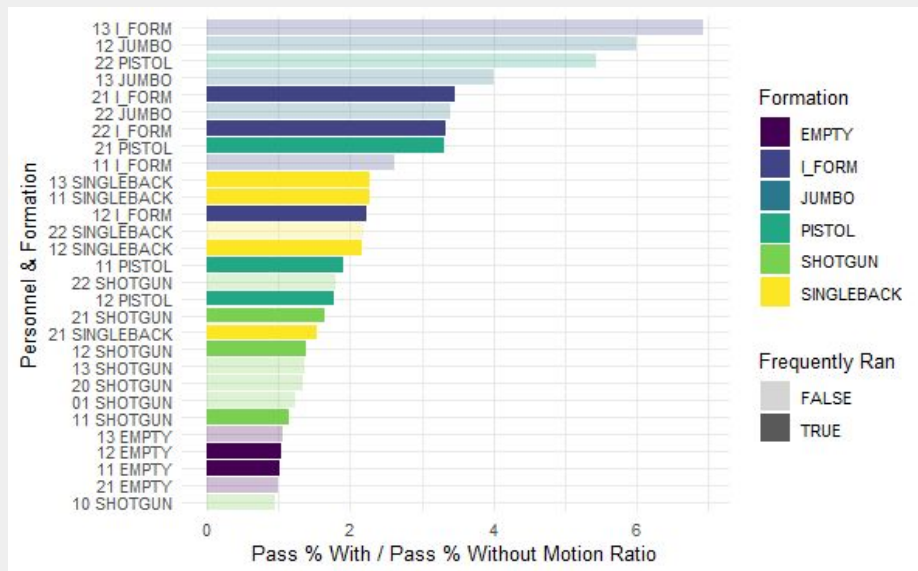
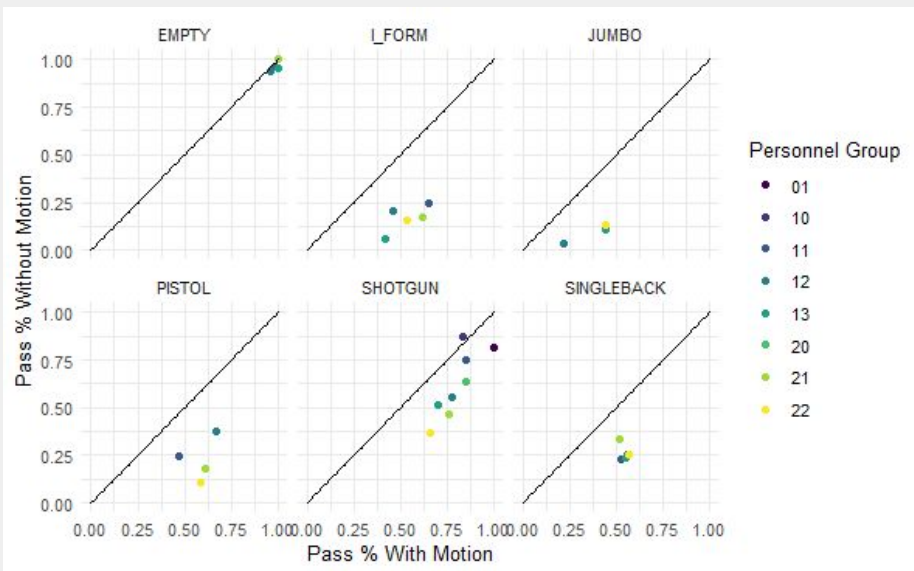
INITIAL OBSERVATIONS:

Broad trends in the league, across all packages



PACKAGE TREND ANALYSIS:

Specific data by package



SO WHAT?

NFL defensive coordinators and defensive players can use this data to inform play style- avoiding falling for play action, making coverage adjustments, etc. This analysis can very easily be extended to other trends, such as predicting PA, pass target location or run direction, and more.